

DARSH SONI

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EDUCATION

B.Tech in Computer Science Engineering (AI & Robotics)

Vellore Institute of Technology, Chennai | 2024 - 2028

CGPA: 8.24/10 (4th sem ongoing)

WORK EXPERIENCE

AI Research Intern

May – Aug 2026

Centre of Cyber Security and Physical Systems, VIT Chennai

- Developing a glacier displacement estimation and prediction pipeline using Sentinel 1 SAR data, DInSAR processing, and ML-based forecasting models.
- Making preprocessing and temporal analysis workflows for horizontal and vertical glacier displacement visualization using Google Earth Engine and Python.

Space Tech Intern

Jan – Mar 2026

Bharat Space Education Research Centre (BSERC), Indian Space Labs, IIT Madras

- Worked on drone systems, PCB prototyping, GIS workflows, and remote sensing applications.
- Studied CanSat/CubeSat systems, telemetry, and rocketry technologies.

Google Student Ambassador (GSA)

Aug – Dec 2025

VIT Chennai (Sole Representative)

- Served as Google's student liaison for 34,000+ students at VIT Chennai and conducted GenAI workshops.

Open Source Contributor

Jul – Oct 2025

Girl Script Summer of Code (GSSoC)

- Contributed 24+ merged pull requests across open-source web and software projects; placed in top 1%.

Team Lead

Jul – Oct 2025

French Club, VIT Chennai

- Led planning and coordination of club events, competitions, and international workshops.

RESEARCH & TECHNICAL PROJECTS

Glacier Horizontal and Vertical Displacement Prediction using DInSAR and ML

Google Earth Engine, Sentinel-1 SAR, DInSAR, Python, NumPy, ML | **Research** (Ongoing)

- Developing a glacier displacement prediction pipeline using Sentinel 1 SAR imagery, DInSAR processing, and making a ML displacement estimation and prediction model.
- Building preprocessing and temporal analysis workflows with visualisation of horizontal and vertical glacier movement using Google Earth Engine and Python.

Multi-Stage NL to ER Diagram & SQL Generation System

[Live](#) | [GitHub](#)

Google ADK, Agentic AI, Multi-Agent Systems | **Research** (Under IEEE Examination)

- Developed an agentic AI framework to convert natural language into ER diagrams and multiple language SQL schemas like Oracle SQL, MySQL, SQLite, PostgreSQL.
- Implementing multi-stage workflows involving ambiguity detection, schema refinement, and SQL generation.

AI Based Railway Crack & Obstacle Detection System

[GitHub](#)

Hardware Prototyping, OpenCV, YOLO | **Patent Application Filed** (Under Examination)

- Designed and developed a hardware prototype using MPU6050, IR, and ultrasonic sensors for railway crack, obstacle, and vibration detection.
- Developed computer vision models for railway crack detection and obstacle-type classification using OpenCV and YOLO; contributed to patent documentation for intelligent railway monitoring.

Darsh Archive: Personal Portfolio

[Live](#)

SvelteKit, MERN, UI/UX Design

- Designed and developed a fully responsive retro-inspired developer portfolio using SvelteKit framework.
- Implemented dynamic routing, multilingual support, custom UI components, and interactive UI.

Oasis: Smart IoT Water Cooler System

[GitHub](#)

Multi Sensor, IoT, Embedded Systems, Firebase, Live Monitoring

- Developed an IoT-enabled smart water cooler system with TDS monitoring, leak detection, and water level tracking.
- Integrated sensor-based monitoring with Firebase for real-time data visualization and alerts.
- Built as a prototype for the Ideathon Hackathon with focus on water conservation and monitoring.

TECHNICAL SKILLS

- **Languages:** C, C++, Python, Java, JavaScript
- **AI / ML:** NumPy, Pandas, OpenCV, PyTorch, CNNs, Computer Vision, Data Visualization
- **Web & Frameworks:** React, Node.js, Express.js, SvelteKit, REST APIs, Firebase
- **Databases:** Oracle SQL, MySQL, MongoDB
- **Tools:** Git, Docker, MATLAB, Embedded Systems

ACHIEVEMENTS & CERTIFICATIONS

Hackathons

- Secured 3rd place among 180+ teams in TechnoVIT CloudOps Hackathon.
- Secured 2nd place among 50+ teams in MediCode Hackathon.
- Finalist/Top 20 in 5 additional national-level hackathons.

Certifications

- AWS Solutions Architecture Virtual Experience Program - Deloitte
- Data Analytics Virtual Experience Program - Deloitte
- Google Gemini Certified Educator - Google
- Agentic AI Certification - Jenny Lectures